Dr. Steven Alexander Wright

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Personal Details

Research Interests

Performance Analysis and Optimisation

- Design and development of low-overhead performance analysis tools
- Novel performance models and modelling techniques
- Evaluation of new programming models and novel hardware
- Quantifying performance portability

Parallel File Systems

- Analysis of parallel file systems and bottlenecks
- Development of portable file structures and data structures
- Replication of sensitive I/O patterns in open environments
- Optimisation of parallel file systems on many-user systems

Energy-aware Computing

- Multi-objective optimisation metrics for energy-aware computing
- Heuristic performance models for energy/power
- Reconfigurable computing for computational science

Appointments

Reader in Computer Science – University of York *September 2026 – Present*
Senior Lecturer in Computer Science – University of York *December 2022 – August 2026*
Lecturer in Computer Science – University of York *July 2018 – December 2022*

- My research at York has been primarily focussed on developing and evaluating approaches to building performance portable plasma physics applications. This research has been supported by five funded projects with the UK Atomic Energy Authority and one project funded by EPSRC. In these projects I have been actively collaborating with the York Plasma Institute, the Centre for Fusion Space and Astrophysics, and many of the US Department of Energy laboratories.
- In my time at York I have taught across the entire degree, from first year Theory and Programming modules, through to the M-level modules I have led. For three years I was the module organiser for the Software Testing (SOTE) module. In 2021/22, I developed a new High-Performance Parallel and Distributed Systems (HIPC) module. I was the coordinator for on-campus undergraduate and postgraduate project modules from 2019 to 2022; I was the programme lead for York's Undergraduate programmes from 2022 to 2024. I currently serve as the Chair of the Board of Examiners.

Postdoctoral Research Fellow – University of Warwick *October 2015 – July 2018*

- Between 2015 and 2018 I was a Research Fellow at the University of Warwick working within the Centre for Computation Plasma Physics (CCPP), in collaboration with the Centre for Fusion Space and Astrophysics, UK AWE and Sandia National Laboratories. With CFSA I was working on developing a mini-application representative of the Odin Arbitrary Lagrangian-Eulerian application to enable rapid evaluation of competing technologies. With AWE and SNL, I was working on the development of a new particle-in-cell application (EMPIRE-PIC).

Visiting Research Fellow – Sandia National Laboratories *June 2016 – August 2016*

- While working as a Research Fellow at Warwick, I was seconded to Sandia National Laboratories in order to lead Warwick efforts in aiding the development of a hybrid fluid/particle-in-cell application. Specifically, I was working to understand the performance issues present in parallel particle-in-cell applications and investigating the use of alternative programming models (Kokkos, Asynchronous Multi-tasking, etc.) to mitigate them.

Lecturer – University of Warwick *September 2014 – July 2018*

- Throughout my postdoctoral career at Warwick, I was additionally employed as a Lecturer to teach the CS118 Programming for Computer Scientists module. This was the first module undergraduate Computer Scientists were taught, and consequently this was the largest module in terms of student numbers and contact hours.

Postdoctoral Research Fellow – University of Warwick *April 2014 – October 2015*

- After finishing my Ph. D., I was a Research Fellow funded by an FP7 grant with Rolls Royce, researching the development of performance models from closed-source applications. In particular, I worked on the development of tools to instrument the LS-DYNA finite element code at run-time in order to build performance models that can be used to predict the potential execution time of the application.

Education

Ph. D. Computer Science *September 2010 – July 2014*
Department of Computer Science, University of Warwick, United Kingdom
Supervisor: Professor Stephen Jarvis

- **Thesis Title**: Monitoring, Analysis and Optimisation of I/O in Parallel Applications

M. Eng (Hons) Computer Science (1st Class) *October 2006 – August 2010*
Department of Computer Science, University of Warwick, United Kingdom

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Professional Memberships

ACM Special Interest Group in High Performance Computing (SIGHPC) *2020 – Present*
ACM Professional Member *2014 – Present*

IEEE Member *2012 – Present*

Research

Research Grants and Contracts

University Collaboration Frameworks

AWE Centre of Excellence for Scientific Computing *June 2025 – May 2030*
University of Edinburgh, University of York, Queen's University Belfast, University of Warwick, Imperial College London, University of Oxford, University of Bristol
Value: £7,100,000 (AWE Nuclear Security Technologies)

As Principal Investigator

Evaluating Dataflow Architectures for Unstructured Discrete Transport Algorithms *October 2023 – March 2027*
Grant Value: £87,607 (AWE Nuclear Security Technologies)

EPOC++: A Future-proofed Kinetic Simulation Code for Plasma Physics at Exascale *October 2022 – October 2025*
Grant Value: £295,346 (EPSRC)

FM-WP4 Code Structure and Coordination: Software Support Procurement (Project NEPTUNE) *September 2022 – March 2024*
Grant Value: £104,173 (UK Atomic Energy Authority)

FM-WP4 Code Structure and Coordination: Support and Coordination (Project NEPTUNE) *September 2021 – October 2022*
Grant Value: £84,473 (UK Atomic Energy Authority)

FM-WP4 Code Structure and Coordination: Investigate DSL and Code Generation Techniques (Project NEPTUNE) *January 2021 – October 2021*
Grant Value: £79,183 (UK Atomic Energy Authority)

As Co-Investigator

FM-WP2 Development of Plasma Fluid Referent Model via Exploratory Proxyapps (Project NEPTUNE) *January 2021 – June 2022*
Grant Value: £233,046 (UK Atomic Energy Authority)

STEP Tender - Lot 2 *June 2019 – March 2021*
Grant Value: £432,722 (UK Atomic Energy Authority)

As PDRA

Centre in Computational Plasma Physics *February 2014 – March 2020*
Grant Value: £101,879 (AWE Plc)

Electromagnetic Plasma in Radiation Environments *October 2016 – September 2019*
Grant Value: £120,000 (AWE Plc)

CCP Flagship: A Radiation-Hydrodynamic Code for the UK Laser-Plasma Community *March 2015 – March 2018*
Grant Value: £298,021 (EPSRC)

Performance Modelling of HPC Applications *October 2013 – September 2016*
Grant Value: £60,000 (AWE Plc)

Rolls-Royce/University of Warwick Clean Sky 2 Project *October 2013 – September 2015*
Grant Value: £236,250 (EU)

Publications

Journal Papers

[68] C. Crispin-Bailey, P. Thuphairo, J. Austin, **S. A. Wright**, and A. Moulds, "Exploring the Feasibility of Novel 3D Polyhedrally-Tiled Computing Arrays," *IEEE Access*, vol. 13, pp. 159685–159713, Sept. 2025

[67] **S. A. Wright**, C. P. Ridgers, G. R. Mudalige, Z. Lantra, J. Williams, A. Sunderland, H. S. Thorne, and W. Arter, "Developing performance portable plasma edge simulations: A survey," *Computer Physics Communications*, vol. 298, 2024

[66] M. T. Bettencourt, D. A. S. Brown, K. L. Cartwright, E. C. Cyr, C. A. Glusa, P. T. Lin, S. G. Moore, D. A. O. McGregor, R. P. Pawlowski, E. G. Phillips, N. V. Roberts, **S. A. Wright**, S. Maheswaran, J. P. Jones, and S. A. Jarvis, "EMPIRE-PIC: A Performance Portable Unstructured Particle-in-Cell Code," *Communications in Computational Physics*, vol. 30, pp. 1232–1268, August 2021

[65] D. A. S. Brown, M. T. Bettencourt, **S. A. Wright**, S. Maheswaran, J. P. Jones, and S. A. Jarvis, "Higher-Order Particle Representation for Particle-in-Cell Simulations," *Journal of Computational Physics*, vol. 435, p. 22, June 2021

[64] A. M. B. Owenson, **S. A. Wright**, R. A. Bunt, Y. K. Ho, M. J. Street, and S. A. Jarvis, "An Unstructured CFD Mini-Application for the Performance Prediction of a Production CFD Code," *Concurrency and Computation: Practice and Experience*, vol. e5443, pp. 1–14, July 2019

[63] S. I. Roberts, **S. A. Wright**, S. A. Fahmy, and S. A. Jarvis, "The Power-Optimised Software Envelope," *ACM Transactions on Architecture and Code Optimization (TACO)*, vol. 16, pp. 21:1–27, August 2019

[62] T. Law, J. Hancox, **S. A. Wright**, and S. A. Jarvis, "An Algorithm for Computing Short-Range Forces in Molecular Dynamics Simulations with Non-Uniform Particle Densities," *Journal of Parallel and Distributed Computing (JPDC)*, vol. 130, pp. 1–11, August 2019

[61] S. J. Pennycook, S. D. Hammond, **S. A. Wright**, J. A. Herdman, I. Miller, and S. A. Jarvis, "An Investigation of the Performance Portability of OpenCL," *Journal of Parallel and Distributed Computing (JPDC)*, vol. 73, pp. 1439–1450, November 2013

[60] **S. A. Wright**, S. D. Hammond, S. J. Pennycook, R. F. Bird, J. A. Herdman, I. Miller, A. Vadgama, A. H. Bhalerao, and S. A. Jarvis, "Parallel File System Analysis Through Application I/O Tracing," *The Computer Journal*, vol. 56, pp. 141–155, February 2013

[59] S. J. Pennycook, S. D. Hammond, G. R. Mudalige, **S. A. Wright**, and S. A. Jarvis, "On the Acceleration of Wavefront Applications using Distributed Many-Core Architectures," *The Computer Journal*, vol. 55, pp. 138–153, February 2012

Conference Papers

[58] S. P. Pasupuleti and S. A. Wright, "The Performance-Power Frontier: A Model-Driven Approach to Energy-Aware Application Optimisation," in *ACM International Conference on Supercomputing*, (United States), ACM, July 2026

[57] M. Smith, **S. A. Wright**, Z. Lantra, and G. R. Mudalige, "The P3 Explorer: An Open Database of Performance, Portability, and Productivity," in *Proceedings of the 33rd Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP'25)*, Mar. 2025

[56] Z. Lantra, **S. A. Wright**, and G. R. Mudalige, "OP-PIC – An Unstructured-Mesh Particle-in-Cell DSL for Developing Nuclear Fusion Simulations," in *Proceedings of the 53rd International Conference on Parallel Processing (ICPP'24)*, ACM, Aug. 2024

[55] E. J. Threlfall, R. J. Akers, W. Arter, M. Barnes, M. Barton, C. Cantwell, P. Challenor, J. W. S. Cook, P. V. Coveney, T. Dodwell, B. D. Dudson, P. E. Farrell, T. Goffrey, M. Green, S. Guillas, M. Hardman, P. A. Hill, L. Kimpton, C. MacMackin, B. McMillan, D. Moxey, G. R. Mudalige, J. T. Omotani, J. T. Parker, F. Parra Diaz, O. Parry, T. Rees, C. P. Ridgers, W. Saunders, S. J. Sherwin, S. Thorne, J. Williams, **S. A. Wright**, and Y. Yang, "Software for Fusion Reactor Design: ExCALIBUR Project NEPTUNE: Towards Exascale Plasma Edge Simulations," in *Proceedings of the 29th IAEA Fusion Energy Conference*, Oct. 2023

[54] D. Chester, T. Groves, S. D. Hammond, T. R. Law, **S. A. Wright**, R. Smedley-Stevenson, S. A. Fahmy, G. R. Mudalige, and S. Jarvis, "StressBench: A Configurable Full System Network and I/O Benchmark Framework," in *Proceedings of the 26th High Performance Extreme Computing Conference (HPEC 26)*, IEEE Conference Publishing Service, September 2021

[53] D. Truby, **S. A. Wright**, R. Kevis, S. Maheswaran, A. Herdman, and S. A. Jarvis, "BookLeaf: An Unstructured Hydrodynamics Mini-application," in *Proceedings of the IEEE International Conference on Cluster Computing (CLUSTER 2018)*, IEEE Conference Publishing Service, September 2018

[52] A. M. B. Owenson, **S. A. Wright**, S. A. Jarvis, R. A. Bunt, Y. K. Ho, and M. Street, "Developing and Using a Geometric Multigrid, Unstructured Mesh Mini-application to Assess the Knights Landing Architecture," in *Proceedings of the 26th Euromicro International Conference on Parallel, Distributed and Network-based Processing*, IEEE Conference Publishing Service, March 2018

[51] S. Roberts, **S. A. Wright**, S. Fahmy, and S. A. Jarvis, "Metrics for Energy-Aware Software Optimisation," *Lecture Notes in Computer Science (LNCS)*, vol. 10266, June 2017

[50] J. Dickson, **S. A. Wright**, D. Harris, S. Maheswaran, J. A. Herdman, M. C. Miller, and S. A. Jarvis, "Enabling Portable I/O Analysis of Commercially Sensitive HPC Applications Through Workload Replication," in *Proceedings of the 39th Cray User Group (CUG'17)*, (Redmond, WA), IEEE Computer Society, Los Alamitos, CA, May 2017

[49] R. A. Bunt, **S. A. Wright**, S. A. Jarvis, M. Street, and Y. K. Ho, "Predictive Evaluation of Partitioning Algorithms Through Runtime Modelling," in *Proceedings of the 23rd High Performance Computing, Data, and Analytics (HiPC'16)*, IEEE, December 2016

[48] T. R. Law, J. Hancox, T. M. K. Cheng, R. A. G. Chaleil, **S. A. Wright**, P. A. Bates, and S. A. Jarvis, "Optimisation of a Molecular Dynamics Simulation of Chromosome Condensation," in *Proceedings*

of the 28th International Symposium on Computer Architecture and High Performance Computing (SBAC-PAD), IEEE, August 2016

[47] S. I. Roberts, **S. A. Wright**, D. Lecomber, C. January, J. Byrd, X. Oro, and S. A. Jarvis, “POSE: A Mathematical and Visual Modelling Tool to Guide Energy Aware Code Optimisation,” in *Proceedings of the 6th International Green and Sustainable Computing Conference (IGSC’15)*, IEEE, September 2015

Posters and Workshop Papers

[46] T. Goffrey, K. Bennett, T. Arber, S. Morris, B. Gosling, **S. A. Wright**, A. Naden, S. Bulut, and C. P. Ridgers, “Development of the EPOCH and EPOCpp Codes,” in *High Power Laser Christmas Meeting (HPL’25)*, Dec. 2025

[45] M. A. Smith and **S. A. Wright**, “Investigating Performance, Portability, and Productivity with a Simple Computational Fluid Dynamics Solver,” in *Proceedings of the 41st Annual UK Performance Engineering Workshop (UKPEW’25)*, (Newcastle, UK), Newcastle, UK, December 2025

[44] Z. Storkey, **S. A. Wright**, and I. Gray, “Experiences of Porting Structured and Unstructured Stencil Applications to FPGA using SYCL,” in *SC Workshops 25*, International Conference for High Performance Computing, Networking, Storage and Analysis, pp. 1496–1501, ACM, Nov. 2025

[43] K. Bennett, S. Morris, T. Goffrey, T. Arber, **S. A. Wright**, A. Naden, and S. Bulut, “Re-engineering the EPOCH PIC code in C++,” in *High Power Laser Christmas Meeting (HPL’24)*, Dec. 2024

[42] M. A. Smith and **S. A. Wright**, “Exploring the Performance, Portability, and Productivity of Approaches to Developing Scientific Computing Applications,” in *Proceedings of the 40th Annual UK Performance Engineering Workshop (UKPEW’24)*, (Leeds, UK), Leeds, UK, December 2024

[41] Z. Storkey, **S. A. Wright**, and I. Gray, “Early Experiences with SYCL on Intel FPGAs: Porting a Heat Diffusion Mini-Application,” in *Proceedings of the 40th Annual UK Performance Engineering Workshop (UKPEW’24)*, (Leeds, UK), Leeds, UK, December 2024

[40] M. A. Smith, **S. A. Wright**, Z. Lantra, and G. R. Mudalige, “The P3 Explorer: Exploring the Performance, Portability, and Productivity Wilderness,” in *Proceedings of the ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC’24)*, Nov. 2024

[39] **S. A. Wright**, B. D. Dudson, P. A. Hill, D. Dickinson, E. Higgins, G. R. Mudalige, B. McMillan, and T. Goffrey, “Approaches to Performance Portability for Fusion Applications,” in *ExCALIBUR Programme-wide Workshop*, July 2022

[38] S. Bulut and **S. A. Wright**, “Optimizing Write Performance for Checkpointing to Parallel File Systems Using LSM-Trees,” in *SC-W’23: Proceedings of the SC’23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, pp. 492–501, ACM, Nov. 2023

[37] W. R. Shilpage and **S. A. Wright**, “An Investigation into the Performance and Portability of SYCL Compiler Implementations,” in *High Performance Computing - ISC High Performance 2023 International Workshops, Revised Selected Papers* (A. Bienz, M. Weiland, M. Baboulin, and C. Kruse, eds.), Lecture Notes in Computer Science (LNCS), (Germany), pp. 605–619, Aug. 2023

[36] R. O. Kirk, M. Nolten, R. Kevis, T. R. Law, S. Maheswaran, **S. A. Wright**, S. Powell, G. R. Mudalige, and S. Jarvis, “Warwick Data Store: A Data Structure Abstraction Library,” in *Proceedings of the 11th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS’20)*, pp. 71–85, IEEE Conference Publishing Service, November 2020

[35] D. Chester, **S. A. Wright**, S. D. Hammond, T. R. Law, R. Smedley-Stevenson, S. Maheswaran, and S. A. Jarvis, “Full-System Modeling and Simulation: Contributions Towards Coupling Contention and I/O,” in *Proceedings of the Workshop on Modeling & Simulation of Systems and Applications (ModSim 2019)*, IEEE Conference Publishing Service, August 2019

[34] D. R. Truby, C. Bertolli, **S. A. Wright**, G.-T. Bercea, K. O’Brien, and S. A. Jarvis, “Pointers inside lambda closure objects in OpenMP target offload regions,” in *The 5th IEEE/ACM Workshop on the LLVM Compiler Infrastructure in HPC (LLVM-HPC 2018)*, (Dallas, TX), IEEE, November 2019

[33] D. R. Truby, C. Bertolli, **S. A. Wright**, G.-T. Bercea, K. O'Brien, and S. A. Jarvis, "Implicit Mapping of Pointers Inside C++ Lambda Closure Objects in OpenMP Target Offload Regions," in *UK OpenMP Users' Conference*, (St. Catherine's College, Oxford, UK), University of Oxford, May 2018

[32] D. Brown, **S. A. Wright**, and S. A. Jarvis, "Performance of a Second Order Electrostatic Particle-in-Cell Algorithm on Modern Many-core Architectures," in *Proceedings of the 33rd Annual UK Performance Engineering Workshop (UKPEW'17)*, (Newcastle-upon-Tyne, UK), Newcastle University, Newcastle-upon-Tyne, UK, December 2017

[31] D. G. Chester, **S. A. Wright**, and S. A. Jarvis, "Understanding Communication Patterns in HPCG," in *Proceedings of the 33rd Annual UK Performance Engineering Workshop (UKPEW'17)*, (Newcastle-upon-Tyne, UK), Newcastle University, Newcastle-upon-Tyne, UK, December 2017

[30] D. Brown, **S. A. Wright**, M. Bettencourt, and S. A. Jarvis, "Performance of Second Order Particle-in-Cell Methods on Modern Many-core Architectures," in *Computational Plasma Physics Conference*, (York, UK), Institute of Physics, November 2017

[29] R. Kirk, G. R. Mudalige, I. Z. Reguly, **S. A. Wright**, M. Martineau, and S. A. Jarvis, "Achieving Performance Portability for a Heat Conduction Solver Mini-Application on Modern Multi-core Systems," in *Second International Workshop on Representative Applications (WRAp 2017)*, IEEE Computer Society, July 2017

[28] J. Dickson, **S. A. Wright**, S. Maheswaran, J. A. Herdman, M. C. Miller, and S. A. Jarvis, "Replicating HPC I/O workloads with Proxy Applications," in *1st Joint International Workshop on Parallel Data Storage & Data Intensive Scalable Computing Systems (PDSW-DISCS'16)*, IEEE Computer Society, Los Alamitos, CA, November 2016

[27] **S. A. Wright** and S. A. Jarvis, "Quantifying the Effects of Contention on Parallel File Systems," in *Proceedings of the 29th IEEE International Parallel & Distributed Processing Symposium Workshops & PhD Forum (IPDPSW'15)*, (Hyderabad, India), pp. 932–940, IEEE Computer Society, Washington, DC, May 2015

[26] J. Dickson, S. Maheswaran, **S. A. Wright**, J. A. Herdman, and S. A. Jarvis, "MINIO: An I/O Benchmark for Investigating High Level Parallel Libraries," in *Proceedings of the 27th ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC'15)*, IEEE, September 2015

[25] R. F. Bird, S. J. Pennycook, **S. A. Wright**, and S. A. Jarvis, "Towards a Portable and Future-proof Particle-in-Cell Plasma Physics Code," in *Proceedings of the 1st International Workshop on OpenCL (IWOCL'13)*, (Atlanta, GA), Georgia Institute of Technology, GA, May 2013

[24] R. F. Bird, **S. A. Wright**, D. A. Beckingsale, and S. A. Jarvis, "Performance Modelling of Magnetohydrodynamics Codes," *Lecture Notes in Computer Science (LNCS)*, vol. 7587, pp. 197–209, February 2013

[23] **S. A. Wright**, S. J. Pennycook, and S. A. Jarvis, "Towards the Automated Generation of Hard Disk Models Through Physical Geometry Discovery," in *Proceedings of the 3rd International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computing Systems (PMBS'12)*, (Salt Lake City, UT), pp. 1–8, ACM, New York, NY, November 2012

[22] **S. A. Wright**, S. D. Hammond, S. J. Pennycook, I. Miller, J. A. Herdman, and S. A. Jarvis, "LDPLFS: Improving I/O Performance without Application Modification," in *Proceedings of the 26th IEEE International Parallel & Distributed Processing Symposium Workshops & PhD Forum (IPDPSW'12)*, (Shanghai, China), pp. 1352–1359, IEEE Computer Society, Washington, DC, May 2012

[21] **S. A. Wright**, S. D. Hammond, S. J. Pennycook, and S. A. Jarvis, "Light-weight Parallel I/O Analysis at Scale," *Lecture Notes in Computer Science (LNCS)*, vol. 6977, pp. 235–249, October 2011

[20] **S. A. Wright**, S. J. Pennycook, S. D. Hammond, and S. A. Jarvis, "RIOT – A Parallel Input/Output Tracer," in *Proceedings of the 27th Annual UK Performance Engineering Workshop (UKPEW'11)*, (Bradford, UK), pp. 25–39, The University of Bradford, Bradford, UK, July 2011

Books

[19] P. S. Pacheco, M. Malensek, and **S. A. Wright**, *An Introduction to Parallel Programming (Third Edition)*. Morgan Kaufmann, 2026. (In Preparation)

Editorials

[18] **S. A. Wright**, "Special Issue on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems," *Concurrency and Computation: Practice and Experience*, vol. e7165, June 2022

[17] **S. A. Wright**, "Performance Modeling, Benchmarking and Simulation of High Performance Computing Systems," *Future Generation Computer Systems*, vol. 92, pp. 900–902, December 2018

Books Edited

[16] **S. A. Wright**, ed., *Special Issue: Performance Modeling, Benchmarking and Simulation of High-Performance Computing Systems (In Production)*. Parallel Computing, Elsevier, Dec. 2025

[15] **S. A. Wright**, S. Hunold, and S. D. Hammond, eds., *Proceedings of the 16th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'25)*. SC-W'25: Proceedings of the SC'25 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis, ACM, November 2024

[14] **S. A. Wright**, S. Hunold, and S. D. Hammond, eds., *Proceedings of the 15th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'24)*. SC-W'24: Proceedings of the SC'24 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis, IEEE, November 2024

[13] **S. A. Wright**, S. D. Hammond, and S. A. Jarvis, eds., *Proceedings of the 14th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'23)*. SC-W'23: Proceedings of the SC'23 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis, ACM, November 2023

[12] **S. A. Wright**, S. D. Hammond, and S. A. Jarvis, eds., *Proceedings of the 13th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'22)*. IEEE, November 2021

[11] **S. A. Wright**, ed., *Special Issue: Performance Modeling, Benchmarking and Simulation of High-Performance Computing Systems*, vol. 34 of *Concurrency and Computation: Practice and Experience*. Elsevier, September 2022

[10] **S. A. Wright**, S. A. Jarvis, and S. D. Hammond, eds., *Proceedings of the 12th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'21)*. IEEE Technical Community on High Performance Computing (TCHPC), IEEE, November 2021

[9] **S. A. Wright**, S. A. Jarvis, and S. D. Hammond, eds., *Proceedings of the 11th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'20)*. IEEE Technical Community on High Performance Computing (TCHPC), IEEE, November 2020

[8] **S. A. Wright**, ed., *Special Section On Performance Modeling, Benchmarking and Simulation Workshop*, vol. 92 of *Future Generation Computer Systems*. Elsevier, March 2019

[7] **S. A. Wright**, S. A. Jarvis, and S. D. Hammond, eds., *Proceedings of the 10th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'19)*. IEEE Technical Community on High Performance Computing (TCHPC), IEEE, November 2019

[6] **S. A. Wright**, S. A. Jarvis, and S. D. Hammond, eds., *Proceedings of the 9th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'18)*. IEEE Technical Community on High Performance Computing (TCHPC), IEEE, November 2018

[5] S. A. Jarvis, **S. A. Wright**, and S. D. Hammond, eds., *High Performance Computing Systems. Proceedings of the 8th International Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems Workshop (PMBS'17), Denver, CO, USA, November 12, 2017*, vol. 10724 of *Lecture Notes in Computer Science (LNCS)*. Berlin: Springer, 2017

[4] S. A. Jarvis, **S. A. Wright**, and S. D. Hammond, eds., *Proceedings of the 7th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'16)*. ACM Special Interest Group on High Performance Computing (SIGHPC), IEEE, November 2016

[3] S. A. Jarvis, **S. A. Wright**, and S. D. Hammond, eds., *Proceedings of the 6th International Workshop on Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems (PMBS'15)*. ACM Special Interest Group on High Performance Computing (SIGHPC), ACM New York, NY, November 2015

[2] S. A. Jarvis, **S. A. Wright**, and S. D. Hammond, eds., *High Performance Computing Systems. Proceedings of the 5th International Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems Workshop (PMBS'14), New Orleans, LA, USA, November 16, 2014*, vol. 8966 of *Lecture Notes in Computer Science (LNCS)*. Berlin: Springer, 2015

[1] S. A. Jarvis, **S. A. Wright**, and S. D. Hammond, eds., *High Performance Computing Systems. Proceedings of the 4th International Performance Modeling, Benchmarking, and Simulation of High Performance Computing Systems Workshop (PMBS'13), Denver, CO, USA, November 18, 2013*, vol. 8551 of *Lecture Notes in Computer Science (LNCS)*. Berlin: Springer, 2014

Professional Activities

Organising Committees

- **Workshop Chair** – International Workshop on Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems held as part of ACM/IEEE Supercomputing (PMBS 2013-25)
- **Sponsorship Chair** – IEEE International Conference on Cluster Computing (Cluster 2025)
- **Co-ordinator** – Centre for Computational Plasma Physics Symposium (CCPP 2015-2017)
- **Co-ordinator** – UK Manycore Developer Conference 2017 (UKMAC 2017)
- **Steering Committee** – UK Performance Engineering Workshop (UKPEW 2012)

Review Committee

- Elsevier Journal of Parallel and Distributed Computing (JPDC 2025)
- Elsevier Parallel Computing (ParCo 2025)
- IEEE Access (ACCESS 2025)
- ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC 2018, 2024-25)
- Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS 2011-25)
- IEEE International Conference on Cluster Computing (Cluster 2025)

- International European Conference on Parallel and Distributed Computing (EURO-PAR 2025)
- International Workshop on Modeling and Simulation of Parallel and Distributed Systems (MSPDS 2019)
- IEEE Letters of the Computer Society (LOCS 2019)
- UK Performance Engineering Workshop (UKPEW 2012, 2017)
- ACM Transactions on Storage (ACM ToS 2017)
- Special Session on High Performance Computing for Application Benchmarking and Optimization (HPBench 2016)
- Workshop on Programming Language Approaches to Concurrency- and Communication-cEntric Software (PLACES 2016)

Guest Editor

- Parallel Computing (2025, In Production)
- Concurrency and Computation: Practice and Experience (CCPE vol. 34 (20))
- Future Generation Computer Systems (FGCS vol. 92)
- IEEE Technical Consortium on High Performance Computing (TCHPC 2018, 2019, 2020, 2021)
- IEEE/ACM Special Interest Group in High Performance Computing (SIGHPC 2015, 2016)
- Lecture Notes in Computer Science (LNCS 8551, 8966, 10724)

Grant Review Panels

- Engineering and Physical Sciences Research Council, UK Research and Innovation
  - Standard Research Grant, November 2023: Responsive Mode (2025)
  - Living labs to improve the environmental sustainability of research infrastructure (2023)
- U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research
  - DE-FOA-0002482: Integrated Computational and Data Infrastructure for Scientific Discovery (2021)

Invited Talks/Posters

- *FM-WP4 Software Support Procurement* - NEPTUNE Workshop, Abingdon, Oxfordshire, 2024
- *Project NEPTUNE: Developing Performance Portable Plasma Edge Simulations* - Workshop on Data Science Infrastructures, Leeds, 2024
- *Approaches to Developing Performance Portable Scientific Software* - SPF ExCALIBUR Programme-wide Workshop, Bristol, 2024
- *Exploring the Portability of Parallel programming Models for Project NEPTUNE* - UK Performance Engineering Workshop, 2023
- *Approaches to Performance Portability for Fusion Applications* - ExCALIBUR Programme-wide Workshop, Exeter, 2023
- *The NEPTUNE ExCALIBUR Project* - Centre for Computational Plasma Physics Symposium (CCPP 2022)

Teaching and Administration

Appointments

External Examiner for Newcastle University

- ◇ MSc Advanced Computer Science — *August 2022 – Present*
- ◇ MSc Cloud Computing — *August 2022 – Present*
- ◇ MSc Cyber Security — *August 2022 – October 2024*

Teaching Experience

Senior Lecturer in Computer Science – University of York — *December 2022 – Present*

Module	Contact Hours	Assessment	Students	Student Rating
2025/26 HIPC High-Performance Parallel and Distributed Systems (H/M-Level)	10 Units 11 Laboratories 22 Office Hours	Open Assessment	64	3.8/4.0
2024/25 HIPC High-Performance Parallel and Distributed Systems (H/M-Level)	10 Units 11 Laboratories 22 Office Hours	Open Assessment	50	5.0/5.0 [Rank #1]
2023/24 HIPC High-Performance Parallel and Distributed Systems (H/M-Level)	10 Units 11 Laboratories 22 Office Hours	Open Assessment	45	4.5/5.0 [Rank #3]
2022/23 HIPC High-Performance Parallel and Distributed Systems (H/M-Level)	8 Units 8 Laboratories 20 Office Hours	Open Assessment	55	4.3/5.0 [Rank #3]

Lecturer in Computer Science – University of York — *July 2018 – December 2022*

Module	Contact Hours	Assessment	Students	Student Rating
2021/22 HIPC High-Performance Parallel and Distributed Systems (H/M-Level)	8 Units 8 Laboratories 20 Office Hours	Open Assessment	65	4.4/5.0 [Rank #2]
2020/21 SOTE Software Testing (M-Level)	10 Lectures 4 Laboratories 8 Office Hours	Open Assessment	20	
2019/20 SOTE Software Testing (M-Level)	10 Lectures 4 Laboratories 8 Office Hours	Open Assessment	25	
2018/19 SOTE Software Testing (M-Level)	10 Lectures 4 Laboratories 8 Office Hours	Open Assessment	30	

Lecturer – University of Warwick *September 2014 – July 2018*

Module	Contact Hours	Assessment	Students
2017/18 CS118 Programming for Computer Scientists	20 Lectures 10 Laboratories 20 Office Hours	2-hour Examination 2 Assignments 4 Problem Sets	240
2016/17 CS118 Programming for Computer Scientists	20 Lectures 10 Laboratories 20 Office Hours	2-hour Examination 2 Assignments 4 Problem Sets	190
2015/16 CS118 Programming for Computer Scientists	20 Lectures 10 Laboratories 20 Office Hours	2-hour Examination 2 Assignments 4 Problem Sets	140
2014/15 CS118 Programming for Computer Scientists	20 Lectures 10 Laboratories 20 Office Hours	2-hour Examination 2 Assignments	135

Administrative Roles Held

N8 Compute Intensive Research Steering Group *2025 – Present*
Chair of the Board of Examiners *2024 – Present*
Science Faculty Promotions Panel *2023 – Present*
HPC Management Committee *2019 – Present*
Undergraduate Programme Lead *2022 – 2024*
Departmental Exam Checking Panel *2019 – 2023*
Undergraduate and Postgraduate Project Coordinator *2019 – 2022*

Teaching and Citizenship Feedback

Peer Feedback

Comments from External Examiners

- “The meetings are always brilliantly Chaired with information provided in advance. I am very grateful for the way in which things are administered and the respect that is shown towards me.” (2025, Prof. Graham Braithwaite [Cranfield], MSc Safety Critical Systems Engineering)
- “I have been impressed with both the standard of assessment and the overall organisation of assessment processes. I believe I have managed to influence some minor improvements, but the assessment processes have always been of a high standard. Exam boards have always been efficient and well organised.” (2025, Prof. Keith Martin [Royal Holloway], MSc Cyber Security)
- “Another area of good practice regards the checks done on the final marks, to ensure that the results are in line with expectation. The exam board has a clear policy regarding scaling which is carefully applied. The exam board meetings are well prepared and run smoothly. All the documentation was available before the board meetings including scripts, marks and reports. The board meetings have a well structured agenda and are run in an efficient and well organised manner.” (2025, Prof. Maribel Fernandez [UCL], BSc Computer Science)
- “Overall, the department maintains very high academic standards. A good effort has been put into maintaining consistency by appropriate paperwork work maintained in relation to moderation, the consideration of mark boundaries and exceptional cases, as well as when graduate teaching assistants are used. The exam board is carried out professionally with due oversight and discussions both at the holistic and detailed levels.” (2025, Dr. Blesson Varghese [St. Andrews], BSc Computer Science)

Teaching Delivery

- ◇ “The lab session was well-structured and effectively delivered. Clear concise instructions were given, providing students with an outline of the tasks ahead. A recap and reflection on the previous lecture was also presented which fed into that lab’s activities. Throughout the lab, Steven engaged actively with students, responding to questions, offering guidance and advice.” (2025, Dr. Mike Freeman)
- ◇ “The delivery was engaging, appropriately paced and perfectly timed. The presentation was clear and contained an interesting variation in material (facts, alternative approaches and real-time code generation and execution). A very good lecture.” (2017, Prof. Graham Martin)
- ◇ “Good style of delivery, making good connection with audience. The slight self-deprecation is endearing and works well to put the students on side of the lecturer.” (2016, Prof. Abhir Bhalerao)

Style and Ambience

- ◇ “A very good and interactive session. The use of the live demo was great and very engaging.” (2024, Dr. Kofi Appiah)
- ◇ “The lecture was presented with enthusiasm and confidence. It was clear that Steven knew the material well, and his relaxed but authoritative style was appreciated by the students.” (2017, Prof. Graham Martin)
- ◇ “Very good ambience. Relaxed delivery with humour. Attentive and responsive audience. Demos are particularly well received.” (2016, Prof. Abhir Bhalerao)

Student Feedback

2025/26

- ◇ “Steven is a wonderful lecturer who clearly knows his material and really knows how to explain it.”
- ◇ “This was an excellent module; likely the strongest I have taken while at UoY. Both lecturers are experts in the field, and genuinely enjoy the topic. They also don’t shy away from saying "I don’t know" when appropriate: a sign of an actual expert!”

2024/25

- ◇ “I was a big fan of the nature of the asynchronous lecture content. I typically really dislike this style of teaching, but this is due to extremely negative experiences from the majority of previous modules. This module proved that, when done well, this style of teaching is effective.”

2023/24

- ◇ “Steven was incredibly helpful during labs, as he is always willing to provide assistance with any issues encountered during the labs or at home”
- ◇ “Well taught, and the passion in all the teaching staff for HIPC and their research is evident and leads to a very engaging and interesting module with lots more to learn if one wishes. Asynchronous content was delivered in a well-organized, straightforward, and easy-to-follow way with clear progression throughout the module. I think this module stands out as a fantastic exception when compared to many others available, both in previous years 1 and 2, and in other year 3 choices.”

2022/23

- ◇ “Very engaging and knowledgeable of the subject, demonstrates a high level of enthusiasm. Also very entertaining during Practical sessions.”

Awards

Warwick Award for Teaching Excellence for Postgraduate Research Students	*May 2013*

Committee Memberships

Member of Departmental Education Committee	*2025 – present*
Member of Undergraduate Teaching Committee	*2022 – present*
Departmental representative on Professional Skills Working Group Science Faculty	*2021 – present*
Member of Research Computing Working Group	*2018 – present*
Staff meetings	*2018 – present*
Member of Board of Studies	*2018 – 2025*

Supervision

Current Students

Matthew Smith	*Started 2025*
Suryachandra Pasupuleti	*Started 2024*
Zadok Storkey	*Started 2023*
Serdar Bulut (Distance, Part-time)	*Started 2022*
Arjen Tamerus (Co-supervisor, Cambridge, Part-time)	*Started 2020*

Postdoctoral Research Assistants

Dr. Andrew Naden	*2023 – 2025*

Visiting Students

Giulio Malenza (University of Turin, Italy)	*2025*

Previous Students (Current Employment)

Dr. Zaman Lantra (PDRA, Warwick)	*Warwick, 2021 – 2025*
Dr. Dean Chester (Simulation Engineer, Cornelis Networks)	*Warwick, 2017 – 2021*
Dr. Dominic Brown (Senior Developer, NVIDIA Corp.)	*Warwick, 2016 – 2020*
David Truby (Senior Software Engineer, ARM)	*Warwick, 2015 – 2019*
Dr. Andrew Owenson (HPC Systems Administrator, University of Oxford)	*Warwick, 2015 – 2019*
Dr. James Dickson (Co-founder, Red Fox Labs)	*Warwick, 2014 – 2018*
Dr. Tim Law (Senior Scientist, AWE Nuclear Security Technologies)	*Warwick, 2013 – 2017*
Dr. Stephen Roberts (Senior Software Engineer, Google)	*Warwick, 2013 – 2017*
Dr. Richard Bunt (Software Engineer, Linaro)	*Warwick, 2012 – 2016*

As External Examiner

Dr. Suneth Ekanayake (Warwick)	*2023*
Mark Klaisoongnoen (Edinburgh)	*2025*